

Introduction/Design Objective:

- The design explores different parameters of Mach-Zehnder Interferometer (MZI) and how they impact the waveguide Free Spectral Range (FSR) and Transmission spectrum.
- The design parameters include path length difference (ΔL), waveguide width (w), incident light polarization (TE, TM) and different bending radii.
- The objective is also to understand the gap between design and fabrication, their origin and impact.

Table 1. Design parameter overview

No	Width (nm)	ΔL (um)	Input pol	Bending radius (um)
1	500	100	TE	5
2	500	50		5
3	500	25		5
4	600	100		5
5	400	100		5
6	500	100	TM	10
7	500	50		10
8	500	25		10
9	500	100		7
10	500	100		5

Theory:

- The principal building block of the MZI is a strip waveguide (width = 500 nm, height = 220 nm), grating couplers and Y branch.
- There are in total of 10 design variations, as listed in Table 1.
- Variation on the waveguide width and polarization influences the effective mode index, group index as well as the electric field.
- Variation on the path length difference (ΔL) impacts the FSR.
- Variation bending radius impacts the waveguide loss at the bending location, which further influences the transmission loss (dB).

Modeling and Simulation:

- Table 2 shows the simulated effective index, group index, FSR for all 10 designs. The result is obtained from Tidy 3D simulation at 1550 nm.

Table 2. Effective index, group index and FSR for designs

No	Width (nm)	ΔL (um)	Input pol	Bending radius (um)	n_{eff}	n_g	FSR (nm)
1	500	100	TE	5	2.512	4.205	5.72
2	500	50		5	2.512	4.205	11.44
3	500	25		5	2.512	4.205	22.88
4	600	100		5	2.566	4.042	5.94
5	400	100		5	2.23	4.376	5.49
6	500	100	TM	10	1.872	4.206	5.63
7	500	50		10	1.872	4.206	11.26
8	500	25		10	1.872	4.206	22.52
9	500	100		7	1.872	4.206	5.63
10	500	100		5	1.872	4.206	5.63

- Figure 1a, 1b, 1c shows the E_y field for the TE polarization WG with strip width of 500 nm, which corresponds to design 1, 2, 3.

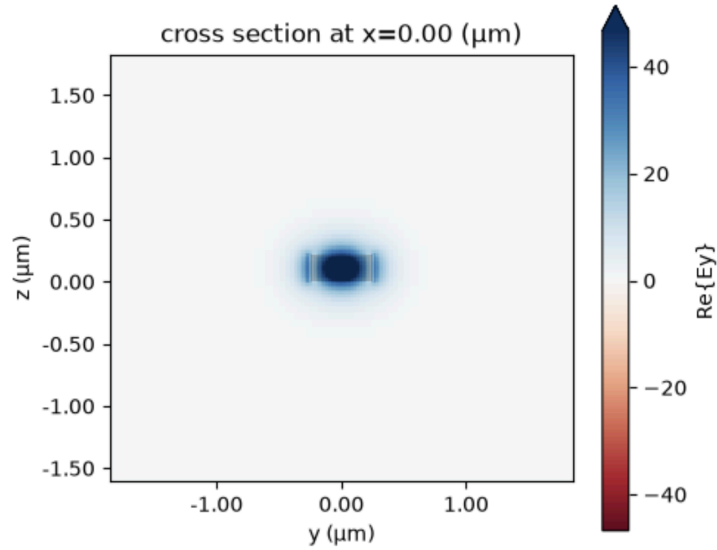


Fig 1a. TE pol, width = 500nm, E_y , 1550 nm

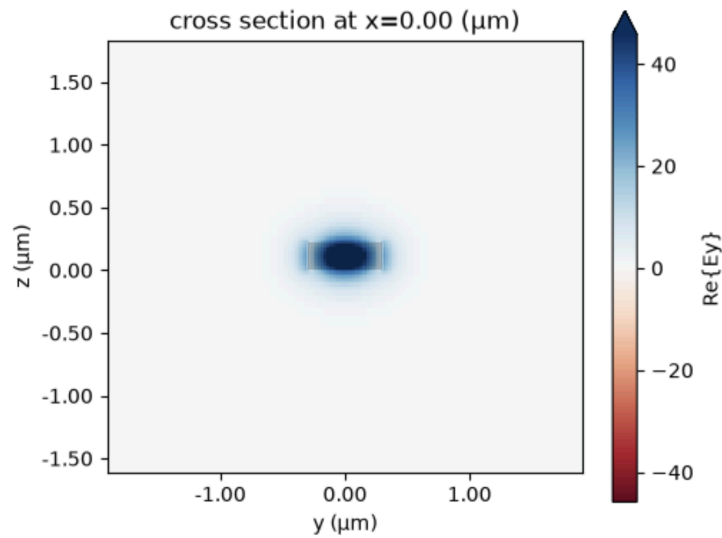


Fig 1b. TE pol, width = 600nm, E_y , 1550 nm

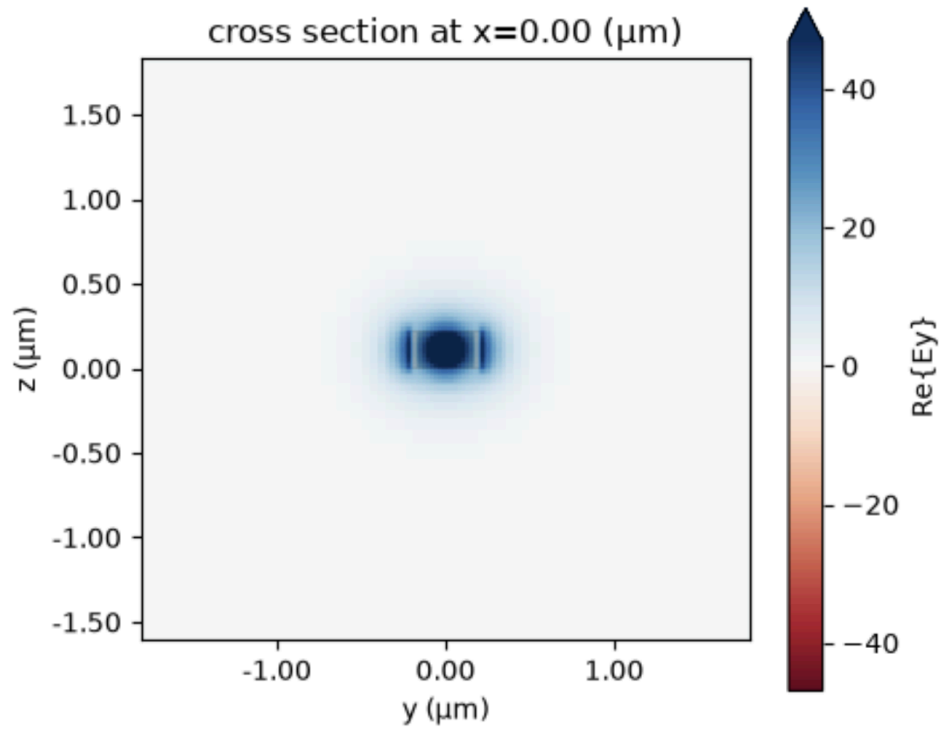


Fig 1c. TE pol, width = 400nm, E_y , 1550 nm

- Figure 2 shows the E_z field of TM pol waveguide with 500 nm width.

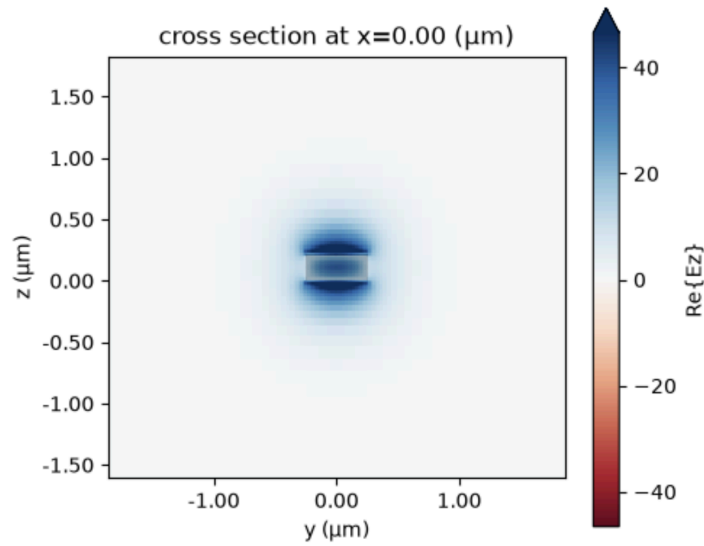


Fig 2. TM pol, width = 500nm, E_z , 1550 nm

- Compact model for the 500 nm width waveguide at 1550 nm, for TE polarization. Simulated in MEEP.

$$n_g(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$$

$$n_g(\lambda) = 2.43 - 1.05(\lambda - 1.55) - 0.134(\lambda - 1.55)^2$$

- Transfer function of the interferometer versus wavelength

$$I_o = I_i/2[1 + \cos(\Delta\beta L)]$$

- Transmission spectrum of the MZI with strip waveguide of 500nm width, path delay of 25 μm , TE mode, 1550 nm, with assumptions of lossless couplers and y branch, no loss at waveguide bending

